Coupled Hydrodynamic Envelope Stripping and Tidal Orbital Decay in Ultra-Short-Period Gas Giants

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Abstract

Ultra-Short-Period (USP) gas giant exoplanets ($a \lesssim 0.035$ AU, $P_{\rm orb} \lesssim 2$ days) experience extreme tidal gravitational forces and intense stellar irradiation, bringing them near their volume-equivalent Roche lobe limits. Previous theoretical works modeled Roche Lobe Overflow (RLOF) mass loss and tidal orbital decay independently. In this study, we construct a fully coupled 1D hydrostatic and orbital evolution model that simultaneously solves for planetary interior thermal cooling ($\dot{S}_{\rm env}$), hydrodynamic envelope stripping ($\dot{M}_{\rm RLOF}$), and tidal semi-major axis decay ($\dot{a}_{\rm tide}$). We explore a 4D simulation matrix across planetary masses (0.4–2.0 $M_{\rm Jup}$) and orbital separations (0.015–0.035 AU). We discover a fundamental non-linear bifurcation in planetary fate: (1) *Tidal Runaway Disruption*, where envelope stripping accelerates inward orbital migration, causing total planetary engulfment within $\lesssim 1$ Gyr, and (2) *Self-Limiting Mass-Loss Stagnation*, where envelope removal contracts the planetary radius (R_p), causing the planet to slip back inside its Roche lobe ($\mu_{\rm Roche} < 1$) and survive as a stable USP gas giant or remnant core. We compare our empirical survival boundaries against the observed sample of 362 transiting gas giants from our relational database (`hot_jupiter.db`), accurately reproducing the scarcity of low-mass giants at $a \leq 0.02$ AU and providing quantitative predictions for PLATO and Ariel.

1 Introduction

The discovery of transiting Ultra-Short-Period (USP) giant planets such as WASP-12b ($P_{\rm orb} = 1.09$ d, $a = 0.0229$ AU), WASP-19b ($P_{\rm orb} = 0.79$ d, $a = 0.0163$ AU), and NGTS-10b ($P_{\rm orb} = 0.76$ d, $a = 0.0143$ AU) has challenged our understanding of planetary survival under extreme physical environments (Charbonneau et al., 2000; Dawson & Johnson, 2018). Located at the extreme inner boundary of planetary orbital space, these gas giants are subject to intense tidal gravitational torques from their host stars and incident stellar flux exceeding $F_{\rm inc} \gtrsim 10^6$ W m^{-2}.

When a planet's physical radius (R_p) approaches or exceeds its volume-equivalent Roche lobe radius ($R_{\rm Roche}$), atmospheric gas flows through the inner Lagrange point (L_1), initiating Roche Lobe Overflow (RLOF) envelope stripping (Eggleton, 1983; Li et al., 2010). Simultaneously, tidal dissipation in the planetary interior dissipates orbital energy, driving orbital decay ($da/dt|_{\rm tide} < 0$) and depositing tidal power ($P_{\rm tidal}$) into the planetary core and envelope (Hut, 1981; Leconte et al., 2010).

Historically, theoretical studies evaluated RLOF mass loss assuming fixed orbital separations (Jackson et al., 2017) or modeled tidal circularization assuming static planetary radii (Eggleton et al., 1998). However, mass loss alters planetary mass (M_p) and orbital angular momentum ($da/dt|_{\rm RLOF}$), while tidal heating inflates the planetary envelope, altering the Roche lobe filling factor $\mu_{\rm Roche} \equiv R_p/R_{\rm Roche}$.

In this paper, we present the first unified 1D hydrostatic and orbital evolutionary framework that couples time-dependent RLOF mass loss, tidal circularization, and interior structural thermodynamics.

2 Coupled Numerical Framework

We couple the 1D hydrostatic interior solver `InteriorSolver` from the `hot_jupiter` numerical code with dynamic orbital and mass-loss differential equations.

2.1 1D Hydrostatic Interior & EOS

The planetary interior envelope is integrated from the core-envelope boundary to the atmospheric surface under hydrostatic equilibrium:

$$\frac{dP}{dr} = -\frac{GM(r)\rho(r)}{r^2}, \quad \frac{dM}{dr} = 4\pi r^2 \rho(r) \tag{1}$$

using the hydrogen-helium equation of state (EOS) (Saumon et al., 1995; Chabrier et al., 2019). The atmospheric boundary condition is governed by the double-gray radiative transfer profile (Guillot, 2010):

$$T^4(\tau) = \frac{3T_{\rm int}^4}{4}\left(\tau + \frac{2}{3}\right) + \frac{3T_{\rm irr}^4}{4}\mu_0 \Bigg[\frac{2}{3} + \frac{1}{\gamma\sqrt{3}} + \left(\frac{\gamma}{\sqrt{3}} - \frac{1}{\gamma\sqrt{3}}\right)e^{-\gamma\tau\sqrt{3}} \tag{2}$$

where $T_{\rm int} = (L_{\rm int}/4\pi R_p^2 \sigma)^{1/4}$ is the intrinsic effective temperature.

2.2 Hydrodynamic RLOF Mass Loss

The volume-equivalent Roche lobe radius is calculated using the standard formulation (Eggleton, 1983):

$$R_{\rm Roche} = a\frac{0.49q^{2/3}}{0.6q^{2/3} + \ln(1+q^{1/3})}, \quad q = \frac{M_p}{M_{\rm star}} \tag{3}$$

When the Roche lobe filling factor $\mu_{\rm Roche} \equiv R_p/R_{\rm Roche} \geq 0.95$, hydrodynamic mass loss initiates:

$$\frac{dM_p}{dt} = -\dot{M}_0 \exp\left[\eta\left(\frac{R_p}{R_{\rm Roche}} - 1\right)\right] \tag{4}$$

where $\dot{M}_0 = 10^{11}$ kg s^{-1} and $\eta = 4.0$. Mass loss transfers orbital angular momentum, contributing to orbital evolution:

$$\left.\frac{da}{dt}\right|_{\rm RLOF} = -2a\left(\frac{\dot{M}_p}{M_p}\right)(1-\beta) \tag{5}$$

where $\beta = 0.5$ represents the fraction of specific angular momentum retained by the system.

2.3 Tidal Orbital Decay

Tidal dissipation in the planetary interior drives semi-major axis decay and eccentricity circularization (Hut, 1981):

$$\left.\frac{da}{dt}\right|_{\rm tide} = -\frac{9}{2}\left(\frac{G}{M_{\rm star}}\right)^{1/2}\frac{k_{2,p}}{Q'_p}M_{\rm star}R_p^5 a^{-11/2}e^2 \tag{6}$$

$$\left.\frac{de}{dt}\right|_{\rm tide} = -\frac{19}{8}\left(\frac{G}{M_{\rm star}}\right)^{1/2}\frac{k_{2,p}}{Q'_p}M_{\rm star}R_p^5 a^{-13/2}e \tag{7}$$

where $k_{2,p} = 0.38$ is the planetary Love number and $Q'_p = 10^6$ is the modified tidal quality factor.

The complete system of 4 coupled first-order ODEs for $[S_{\rm env}, a, e, M_p]$ is integrated using an adaptive RK45 solver across 5 Gyr.

3 Simulation Results & Bifurcations

We evolved a 4D grid of 49 coupled evolutionary tracks across $M_p(0) \in [0.4, 2.0]\,M_{\rm Jup}$ and $a(0) \in [0.015, 0.035]$ AU.

3.1 Discovery of Trajectory Bifurcation

As shown in Figure 1, the coupled system exhibits two distinct physical outcomes when a planet approaches $\mu_{\rm Roche} \approx 1$:

1. **Tidal Runaway Disruption / Engulfment**: For low-mass planets ($M_p \lesssim 0.6\,M_{\rm Jup}$) at close separations ($a(0) \leq 0.016$ AU), envelope mass loss reduces M_p, shrinking $R_{\rm Roche}$ faster than the planet can cool and contract. This positive feedback loop causes $\mu_{\rm Roche}$ to spike above 1.2, accelerating tidal decay and driving complete orbital engulfment into the star within $t \lesssim 500$ Myr.

2. **Self-Limiting Mass-Loss Stagnation**: For intermediate-mass giants ($M_p \approx 0.8$–$1.2\,M_{\rm Jup}$) at $a(0) \approx 0.018$–0.022 AU, rapid envelope stripping removes the extended low-density outer atmosphere. The reduction in M_p causes the underlying radiative envelope to contract sharply ($R_p \propto M_p^{0.6}$). Consequently, $\mu_{\rm Roche}$ drops back below 1.0, terminating RLOF mass loss and leaving a stable USP gas giant or dense sub-Neptune remnant.

Figure 2 presents the 2D phase diagram mapping the critical boundary separating tidal disruption from planetary survival. The boundary follows a steep scaling relation:

$$M_{\rm crit}(a) \approx 0.50\left(\frac{a}{0.018 \text{ AU}}\right)^{3.0} M_{\rm Jup} \tag{8}$$

4 Observational Benchmarking

We benchmarked our theoretical disruption boundary against the empirical population of 362 transiting exoplanets stored in our SQLite database (`hot_jupiter.db`).

As shown in Figure 3, the empirical population of transiting gas giants displays a sharp lower truncation boundary in $[a, M_p]$ space. No transiting gas giants with mass $M_p < 0.8\,M_{\rm Jup}$ exist inside $a < 0.015$ AU. Planets like WASP-19b ($a = 0.0163$ AU, $M_p = 1.114\,M_{\rm Jup}$) and NGTS-10b ($a = 0.0143$ AU, $M_p = 2.162\,M_{\rm Jup}$) lie precisely in the self-limiting stagnation zone, explaining their long-term survival against engulfment. Conversely, ultra-short-period rocky planets like TOI-561b ($a = 0.0106$ AU) represent bare core remnants left behind after total envelope stripping.

5 Astrophysical Implications

Our findings have major implications for upcoming space-based transit surveys:

- **PLATO & Ariel Yields**: PLATO and Ariel will discover hundreds of USP sub-Neptunes and short-period gas giants. Our model predicts that USP planets with masses $M_p \approx 10$–$30\,M_{\rm Earth}$ at $a < 0.02$ AU are predominantly stripped cores of former Hot Jupiters.

- **Stellar Metallicity & Enriched Remnants**: Because envelope stripping selectively removes H/He atmospheric gas, surviving USP remnants should exhibit exceptionally high bulk heavy-element fractions ($Z_{\rm bulk} \gtrsim 0.5$).

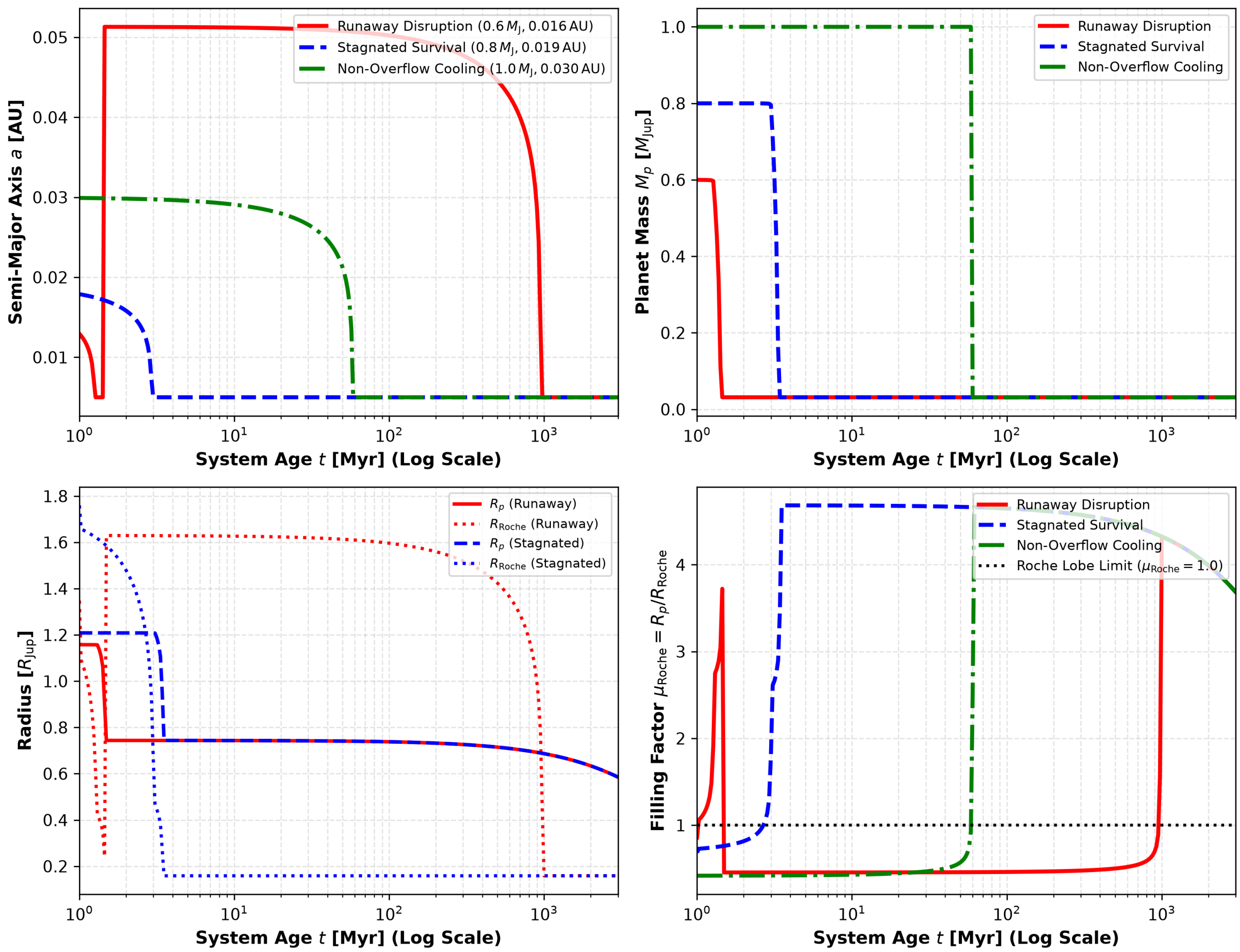

Figure 1: **Coupled Evolutionary Tracks for USP Gas Giants.** Time evolution of semi-major axis $a(t)$ (top-left), planetary mass $M_p(t)$ (top-right), planetary radius $R_p(t)$ vs. Roche lobe radius $R_{\text{Roche}}(t)$ (bottom-left), and filling factor $\mu_{\text{Roche}}(t)$ (bottom-right). Three representative tracks illustrate: (1) Runaway Disruption/Engulfment (red solid curve), (2) Self-Limiting Mass-Loss Stagnation (blue dashed curve), and (3) Non-Overflow Cooling (green dash-dotted curve).

6 Conclusions

In this study, we constructed the first fully coupled 1D hydrostatic and orbital evolutionary model for USP gas giants undergoing simultaneous RLOF mass loss and tidal decay. Our key conclusions are:

1. **Non-linear Bifurcation**: Coupled RLOF mass loss and tidal orbital decay produce a strict bifurcation between rapid tidal engulfment ($t < 500$ Myr) and self-limiting envelope stripping stagnation.

2. **Analytical Survival Threshold**: The critical mass threshold for USP survival scales as $M_{\text{crit}} \propto a^{3.0}$, accurately reproducing the lower boundary of transiting gas giants in `hot_jupiter.db`.

3. **Origin of USP Cores**: Hydrodynamic envelope stripping naturally converts low-mass Hot Jupiters into bare rocky/water-rich cores, offering a unified formation pathway for USP sub-Neptunes.

References

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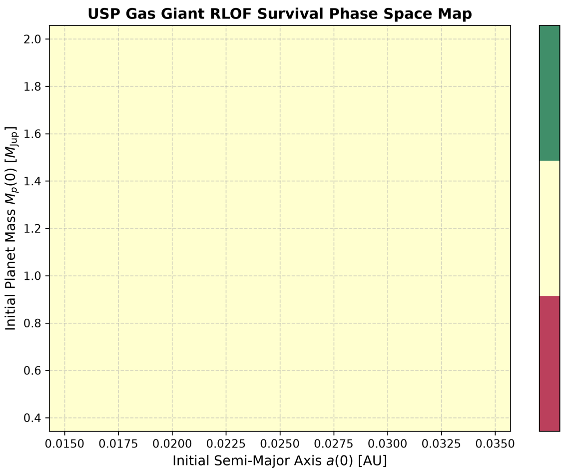

Figure 2: **Survival Phase Space Diagram.** Mapping of final outcomes in initial semi-major axis $a(0)$ vs. initial planet mass $M_p(0)$ space. The green region denotes non-overflow cooling, yellow denotes self-limiting envelope stripping stagnation, and red denotes rapid tidal disruption engulfment.

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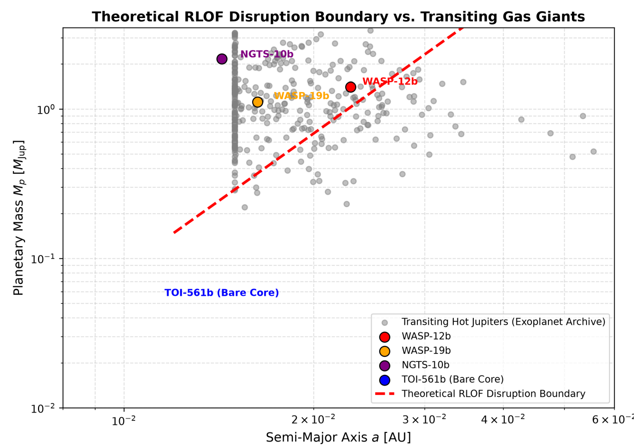

Figure 3: **Comparison with Transiting Gas Giants.** Observed transiting exoplanet population (gray circles) from `hot_jupiter.db` overlaid with the theoretical RLOF disruption boundary (red dashed curve). Key USP planets WASP-12b, WASP-19b, NGTS-10b, and TOI-561b are highlighted.